BUSINESS REQUIREMENTS DOCUMENT (BRD)

MEDITRACK NIGERIA

Project Name: MediTrack Nigeria
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Prepared By: Business Analyst
Prepared For: Medical Director, Doctors,Federal Ministry of Health , Lagos state ministry of Health, Hospital IT department
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1. EXECUTIVE SUMMARY

1.1 Background

Nigerian public hospital patients spend an average of 4.5 hours waiting to see a doctor for a consultation that lasts only 10 minutes. If MediTrack reduces this waiting time to 90 minutes, the country's 85 million annual outpatient visits could restore approximately 255 million productive hours to Nigerians every year. To address this challenge, we recommend the deployment of MediTrack Nigeria at the proposed 200-bed pilot hospital in Lagos State. MediTrack is a digital patient management system that combines appointment booking, real time queue management, digital patient records, digital payment processing, and an operational dashboard into a single platform. By replacing the current paper-based process with a more efficient digital solution, the hospital will be better positioned to deliver faster, safer, and more reliable healthcare services.

The first reason for this recommendation is that MediTrack will significantly reduce patient waiting time and improve service delivery. Patients will be able to book appointments through either a web portal or a USSD platform, eliminating the need to arrive at the hospital before dawn to secure a place in the queue. The real-time queue management system will also improve patient flow across departments, reducing congestion and making healthcare services more accessible and organised.
Secondly, MediTrack will improve the quality of patient care through secure digital patient records. The current paper-based system records a patient folder missing rate of approximately 22–28%, often forcing doctors to consult patients without access to their medical history. With digital records that can be accessed instantly by authorised healthcare professionals, patient information will be readily available, supporting more accurate diagnoses, better treatment decisions, and continuity of care.
Finally, MediTrack will strengthen financial accountability and operational decision-making. Manual payment processes currently contribute to an estimated 18–25% revenue leakage through under-billing, missing charges, and reconciliation errors. The proposed system introduces digital payment receipts, automated SMS confirmations, and a live management dashboard that provides real-time insights into patient volumes, queue lengths, ward occupancy, and hospital performance. These features will improve revenue recovery, increase transparency, and enable management to make timely, data-driven decisions.
The purpose of this document is to summarise the findings of the proposed MediTrack implementation at the pilot hospital and provide decision-makers with sufficient information to approve the deployment of an integrated digital hospital management system both at the pilot hospital

and upon successful implementation, a national roll-out, that addresses the root causes of excessive patient waiting times, missing medical records, revenue leakages and operational bottlenecks in Nigerian public hospitals.

2. PROBLEM STATEMENT

Element	Operational Context	Key metric
Situation *(Current Reality)*	Patients face extreme arrival-to-consultation delays for brief medical appointments. Clinical workflows rely entirely on physical paper storage networks, and payment collections are managed through paper-based cashier desks. These legacy administration models are deployed across a massive nationwide footprint.	• 4.5 hours average wait time for a 10-minute appointment • 22–28% of patient folders are missing or unavailable • 18–25% revenue leakage from monthly collections • 85 million outpatient visits across public facilities annually
COMPLICATION *(The Systemic Frictions)*	Core tasks require a physical presence, forcing patients to travel just to book slots, queue for registration, wait for manual folder retrieval, and await cash processing. Furthermore, digital exclusion affects patients with basic phones, while severe infrastructure limits cause the pilot site to lose power for multiple hours daily.	• Zero web/ USSD alternatives for basic/offline feature phone users • 4 hours average daily grid power interruptions at pilot site • 6 manual steps required per patient on-site to see a doctor
IMPACT *(The Compounded Threat)*	Inefficiencies lead to congested facilities and high staff fatigue. Doctors routinely consult patients blindly without critical medical backgrounds, heightening care risks. Unmonitored processing causes major financial leaks, and millions of citizen hours are permanently lost to hospital delays every year.	• Doctors consult without histories due to folder losses • 18–25% unrecovered operational funds needed for facilities • 255 million productive hours stripped from Nigerians annually
QUESTION *(The Core Product Goal)*	How might MediTrack Nigeria provide an accessible, secure, and offline-resilient digital hospital platform that cuts waiting times, streamlines information access, and stops revenue leakage without excluding patients using basic phones?	Target Outcome: Drop wait times to 90 minutes and restore 255 million productive hours annually via an offline-capable,USSD or Web enabled platform.

3. STAKEHOLDER ANALYSIS

Name/Role	Power / Interest	Primary Need	Key Concern	Key Concern	Communication Frequency
Medical Director	High Power	Real-time operational	Disruptions to	Weekly progress	Weekly

(Project Sponsor)	High Interest	visibility via management dashboard.	clinical operations during system rollout.	briefings and hands-on dashboard walk-throughs.	
Federal Ministry of Health *(Funder/Policy Authority)*	High Power High Interest	Clear KPI evidence at Month 12 to justify national roll-out.	Patient data security and compliance with NDPR laws.	Milestone reviews, security audit sign-offs, and KPI reports.	Monthly
Lagos State Ministry of Health *(Co-Funder/State Oversight)*	High Power Medium Interest	State-level reporting data to plan health resource allocation.	Alignment with state healthcare benchmarks and timelines.	Periodic data-sharing workshops and automated export updates.	Quarterly
NHIS *(Regulatory Body)*	High Power Low Interest	Strict compatibility with standard NHIS billing codes.	Liability and compliance issues if billing errors occur.	Alignment meetings during the database configuration phase.	Milestone-Based
Doctors *(12 at Pilot Hospital)*	Medium Power High Interest	Seamless interface to view patient records and queue lists.	Increased documentation time and typing during consultations.	Co-designing zero-typing templates and voice/click alternatives	Bi-weekly
Nurses *(40 at Pilot Hospital)*	Medium Power High Interest	Streamlined tools to manage ward queues and record arrivals	Increased workload and lack of instant technical support.	Appointing super-users per shift and providing a 24/7 helpdesk.	Bi-weekly
Hospital Cashiers & Admin *(8 Staff)*	Low Power High Interest	Efficient, secure digital payment and registration modules.	Job replacement fears and lack of offline paper backups.	Re-skilling as system administrators; designing offline modes.	Weekly
Hospital IT Department *(Ministry Team)*	Medium Power High Interest	Standardized infrastructure to host and maintain the platform.	High post-pilot maintenance burden on a small team	Comprehensive technical handovers and specialized training.	Monthly
Smartphone Patients *(e.g., 32yo Lekki Accountant)*	Low Power High Interest	Remote appointment booking via a fast, modern web portal.	System lag or failing to secure the selected time slot.	Launching a responsive web portal with instant confirmations.	Per Visit
Basic Phone Patients *(e.g., 68yo Badagry Woman)*	Low Power High Interest	Working USSD code (`*456*MEDI#`) with SMS confirmations.	Language barriers and English-only menus.	Deploying multilingual USSD menus (Yoruba, Hausa, Igbo).	Per Visit
Urgent Care Mothers *(Mother with Sick Infant)*	Low Power High Interest	Immediate bypass of standard digital registration paths.	Complicated interfaces and delays during emergencies.	Creating an emergency-override fast-track queue option.	Per Visit
Wage-Earner Patients	Low Power High Interest	Guarantee that arrival matches the actual	Losing a full day's wages to	Sending real-time SMS queue	Per Visit

(e.g., Construction Worker)		consultation time.	unpredictably long queues.	countdowns to prevent early waiting.	

4. PROJECT SCOPE

4.1 In Scope

- USSD appointment booking (*456*MEDI#) for all departments - Core booking channel for the majority of patients who use basic phones.
- Web portal appointment booking for smartphone users.
- Real-time queue management display screens in waiting areas.
- Digital patient registration (phone number + NIN optional; no document required).
- Basic digital visit record (date, department, complaint, doctor, treatment summary).
- Digital payment receipt via SMS within 60 seconds of payment.
- Admin dashboard: live occupancy, average wait time, daily patient count.
- Automated SMS appointment reminders (24 hrs. and 2 hrs. before appointment).
- Multi-language USSD menus (Yoruba, Igbo, Hausa, Pidgin English).
- Walk-in patient registration and queue insertion alongside pre-booked patients.
- Offline-Resilient Operation.

4.2 Out of Scope

OUT OF SCOPE	**Justification**
Full Electronic Health Records with ICD-10/SNOMED clinical coding	Requires a separate clinical system integration and is planned for a later phase.
NHIS national database integration	Requires Ministry-level API approval and a formal data-sharing agreement.
Pharmacy inventory management	Pharmacy inventory is a separate regulatory and supply-chain domain.
Telemedicine video consultation	Requires separate MDCN licensing, infrastructure and additional budget.

5. Functional Requirement

Module Name	Req ID	Functional Requirement
Patient Registration	FR-PR-001	The system shall allow new patients to register using only Nigerian mobile numbers (NIN optional) via USSD menu under 3 minutes.
	FR-PR-002	The system shall generate a unique 8-digit MediTrack Patient ID after successful registration.
	FR-PR-003	The system shall send the Patient ID to the patient's registered phone by SMS within 30 seconds of successful registration.
Appointment Booking — USSD	FR-ABU-004	The system shall allow patients to access appointment booking through **456**MEDI#.
	FR-ABU-005	The system shall display available appointment slots across the next five business days
	FR-ABU-006	The system shall generate an appointment reference after successful booking.
	FR-ABU-007	The system shall send the appointment date, time, department and floor number to the patient by SMS within 30 seconds.
Appointment Booking — Web	FR-ABW-008	The system shall provide a web-based appointment calendar for authorised users.
	FR-ABW-009	The system shall display available appointment slots by department and date.
	FR-ABW-010	The system shall refresh appointment availability within 60 seconds after a booking, cancellation or slot addition.
Queue Management	FR-QM-011	The system shall assign each patient a queue position after registration/check-in.
	FR-QM-012	The system shall display the current queue position on department waiting-area screens.
	FR-QM-013	The system shall update queue information within 10 seconds after a nurse or doctor calls the next patient.
	FR-QM-014	The system shall record when a consultation is marked as completed
Patient Records	FR-PR-015	The system shall create a digital visit record for every completed consultation.
	FR-PR-016	The visit record shall contain the MediTrack Patient ID, date, department, presenting complaint, Doctor ID and treatment summary.
	FR-PR-017	The system shall make an authorised patient's visit record available within 60 seconds after the doctor confirms the record**.**

	FR-PR-018	The system shall restrict patient records to unauthorised users.
Digital Payments	FR-DP-019	The system shall record hospital fee payments digitally.
	FR-DP-020	The system shall generate a unique payment reference for every recorded payment.
	FR-DP-021	The system shall send an SMS payment receipt within 60 seconds after successful payment confirmation including: amount paid, service description, unique payment reference, and hospital name.
Administration Dashboard	FR-AD-022	The system shall display on the Medical Director's dashboard current hospital occupancy per ward (refreshed every 15 minutes).
	FR-AD-023	The system shall display the average patient wait time per department for the day, total patients seen in a day vs the same day last week.
	FR-AD-024	The system shall generate an alert when ward occupancy exceeds 90% capacity.
Notifications	FR-N-025	The system shall send an automated SMS appointment reminder to each booked patient 24 hours and 2 hours before their appointment, including: appointment time, department name, floor number, and any preparation instructions set by the booking clinician and whenever an appointment is cancelled or rescheduled before their appointment time.
	FR-N-026	The system shall notify doctors within 10 seconds whenever laboratory or radiology results are uploaded.

6. NON-FUNCTIONAL REQUIREMENTS

Requirement Category	Req ID	Non-Functional Requirement
Offline Resilience *(The Primary Constraint)*	NFR-001	The system shall operate in offline mode for a minimum of 6 continuous hours using local device cache during a power or internet outage, with automatic synchronization to the central server within 5 minutes of connectivity being restored, and zero data loss from any transaction entered during the offline period.
Performance	NFR-002	The system shall process USSD requests and display the next menu option within 3 seconds for at least 95% of transactions over a 2G mobile network.
	NFR-003	The system shall display appointment availability within 60 seconds after a booking, cancellation or slot addition.
Availability	NFR-004	MediTrack shall maintain at least 99% availability during the hospital's operating period of 6:00 AM–10:00 PM, excluding

		scheduled maintenance.
Security	NFR-005	The system shall encrypt patient personal and health information both at rest and in transit and prevent unauthorised users from accessing patient records.
Usability	NFR-006	A first-time basic-phone user shall be able to complete USSD appointment booking within three minutes without assistance
Scalability	NFR-007	The system architecture shall support expansion from one pilot hospital to at least 50 hospitals without redesigning the core appointment, queue and patient-record architecture.
Data Protection	NFR-008	Patient and transaction records shall be retained for a minimum of five years, subject toCBN/NHIS applicable regulatory requirements.
Multilanguage	NFR-009	USSD menus shall be available in English, Yoruba, Igbo, Hausa and Pidgin English.

7. USER STORIES WITH ACCEPTANCE CRITERIA

US-ID	User story	Acceptance Criteria
US-001	As a patient with a Nokia basic phone, I want to book an appointment via USSD so that I do not have to travel to the hospital and queue from 5:00 AM.	1. Given I have a registered Nigerian mobile number When I dial **456**MEDI# and select a department and available slot Then the system shall confirm my appointment and send the appointment details by SMS. 2. Given no appointment slot is available When I select the department Then the system shall inform me that no slot is currently available. 3. Given I have successfully booked an appointment When the booking is completed Then the system shall provide a unique appointment reference.
US-002	As a doctor, I want to view my next patient's complete visit history before they enter my consultation room so that I can consult without asking them to remember past diagnoses.	1. Given I am an authorised doctor When I open the patient's record Then I shall see the patient's authorised previous visit information. 2. Given the patient has no previous digital visit record

		When I access the record Then the system shall indicate that no previous digital record is available. 3. Given I am not authorised to access the record When I attempt to open it Then access shall be denied.
US-003	As a hospital cashier, I want to record payment digitally so that the system generates and sends a receipt automatically and I cannot accidentally issue a receipt for cash I did not physically receive.	1. Given a payment has been received When I record the transaction Then the system shall generate a unique payment reference. 2. Given the transaction is successfully recorded When payment confirmation is completed Then the patient shall receive an SMS receipt within 60 seconds. 3. Given required payment information is incomplete When I attempt to submit the transaction Then the system shall prevent completion until the missing information is provided.
US-004	As a nurse, I want to see the real-time queue for my ward on a wall screen so that I can call the next patient without walking to the front desk.	1. Given patients are waiting in the department When I access the queue Then I shall see patients in their current queue order. 2. Given a patient has been called When I mark the patient as called Then the queue shall update within 10 seconds. 3. Given a consultation is completed When I mark the patient as completed Then the patient shall be removed from the active queue.
US-005	As the Medical Director, I want a live dashboard showing ward occupancy and average wait time per department so that I can redeploy staff from a quiet department to an overloaded one in real time.	1. Given I am an authorised Medical Director When I open the dashboard Then I shall see current occupancy and average waiting time. 2. Given a ward exceeds 90% capacity When the occupancy reaches the threshold Then the dashboard shall display a red capacity alert. 3. Given today's patient numbers are available When I view the dashboard Then I shall see today's patient count alongside the same-day figure from the previous week.
US-006	As a patient, I want an SMS reminder 2 hours before my appointment so that I do not forget and waste my	1. Given I have a confirmed appointment

	booked slot for someone else.	When 24 hours remain before the appointment Then the system shall send an SMS reminder. 2. Given the appointment is two hours away When the two-hour reminder period is reached Then the system shall send another SMS. 3. Given preparation instructions have been configured When the reminder is sent Then the SMS shall include the applicable instructions.
US-007	As the Medical Director, I want all clinical records to remain accessible during a power outage so that patient care is never interrupted because NEPA took light.	1. Given the hospital loses power or internet connectivity When MediTrack enters offline mode Then core supported functions shall remain available for at least six hours. 2. Given staff enter transactions during offline mode When connectivity returns Then the system shall synchronise the transactions within five minutes. 3. Given transactions were recorded successfully offline When synchronisation occurs Then no successfully recorded transaction shall be lost.
US-008	As an NHIS compliance officer, I want every payment receipt to include a valid NHIS billing code so that I can verify insurance claims against MediTrack transaction records.	1. Given a payment is recorded When the receipt is generated Then the required billing reference shall be included. 2. Given a transaction is completed When an authorised compliance officer searches for it Then the transaction shall be retrievable. 3. Given an unauthorised user attempts to access payment records When access is requested Then the system shall deny access.
US-009	As a nurse, I want the system to confirm patient identity before I update a queue status so that the wrong patient is not marked as attended.	1. Given the nurse selects a patient from the queue, when the patient record is opened then the system shall display identifying information for confirmation. 2. Given the patient's identifying information is displayed, when the nurse confirms that the information matches the patient, then the system shall allow the nurse to update the queue status. 3. Given the nurse does not confirm the patient's

		identity, when the nurse attempts to update the queue status, then the system shall prevent the status from being changed.
US-010	As a hospital administrator, I want unauthorised access attempts recorded so that patient information remains protected.	1. Given a user does not have permission to access a patient's medical record, when the user attempts to access the record, then the system shall deny access. 2. Given an unauthorised access attempt has occurred, when the system denies the request, then the system shall record the user, date, time, and attempted access. 3. Given an unauthorised access attempt has been recorded, when an authorised administrator reviews the security records, then the incident shall be available for review.
US-011	As a patient, I want to cancel an appointment so that the slot can become available to another patient.	1. Given the patient has an upcoming appointment, when the patient selects the cancellation option, then the system shall display the appointment details and request confirmation. 2. Given the patient confirms the cancellation, when the cancellation is submitted, then the system shall change the appointment status to Cancelled. 3. Given an appointment has been cancelled, when the cancellation is completed, then the appointment slot shall become available for another patient to book.
US-012	As a receptionist, I want to add a walk-in patient to the queue so that patients without appointments can still receive care.	1. Given a patient arrives without an appointment, when the receptionist enters the required patient details, then the system shall create a walk-in registration. 2. Given the walk-in patient has been registered, when the receptionist selects the required service or department, then the system shall assign the patient a queue number. 3. Given a queue number has been assigned, when the registration is completed, then the walk-in patient shall appear in the selected service or department queue.
US-014	As a Ministry health official, I want to generate a monthly hospital performance report so that I can monitor service delivery across participating facilities.	1. Given the Ministry health official is authorised to access reports, when the official selects a reporting month, then the system shall generate the corresponding hospital performance data. 2. Given the report has been generated, when the official views the report, then it shall display the agreed performance indicators for participating facilities. 3. Given the monthly performance report is available, when the official selects the download option, then the system shall provide the report

		in a downloadable format.
US-015	As a doctor, I want to record a laboratory referral so that the patient's required investigation is documented.	1. Given that an authorised doctor is viewing a patient's active visit, when the doctor creates a laboratory referral, then the system shall allow the doctor to enter the required laboratory test and referral details. 2. Given that the laboratory referral has been submitted successfully, when the doctor saves the patient's record, then the referral shall be linked to the patient's MediTrack Patient ID and visit. 3. Given the doctor successfully submits a lab referral,When the referral is saved, Then the system sends an electronic notification to the laboratory department queue and a copy of the referral appears in the patient's active orders list.

8. PROCESS OVERVIEW

Current State

Patients → Hospital Arrival → Registration/Appointment Queue →Manual File Retrieval → Doctor Consultation →Manual Lab/Pharmacy Billing →Manual Cashier Payment →Patients

Issues

- Manual processing
- Long queues
- No central appointment management
- Limited patient visibility
- Manual payment processing
- Higher risk of record and billing errors

Future State (With MediTrack)

Patient → USSD/Web Registration & Appointment Booking → Automated SMS Reminders → Hospital Check-in → Electronic Queue Management → Doctor Consultation → Integrated Digital Billing & Payment → Patient

Benefits

- Automated appointment management
- Reduced waiting time
- Electronic queue visibility

- Improved patient-record management
- Faster billing and payment
- Reduced processing errors
- Better operational visibility

9. ASSUMPTIONS

- Patients are assumed to have access to either a mobile phone capable of USSD/SMS communication or an internet-enabled device for web service.
- Hospitals possess smartphones or computers across all departments
- Payment gateway services remain operational.
- Patients possess mobile phones capable of handling USSD codes or SMS communication.

10. RISKS

Risk	**Impact**	**Probability**	**Mitigation**
System downtime	High	Medium	Cloud redundancy
Data Breach	High	Low	Security control
Payment failures	High	Medium	Multiple payment gateways
Low adoption rate	Medium	Medium	Training

11. PROJECT CONSTRAINTS

- Infrastructure: MediTrack shall support essential registration, queue and patient-record operations during up to four hours of daily power interruption.
- Technology: Patient registration must support USSD and web access throughout the implementation.
- Connectivity: The system must support low-bandwidth environments and synchronise data automatically when connectivity is restored.
- Devices: Deployment must use existing hospital computers and tablets without requiring major hardware upgrades.
- Compliance: Patient records and billing data must comply with the Nigeria Data Protection Act (NDPA) and applicable healthcare regulations.
- Payment Integration: Payment services must use approved hospital payment infrastructure without adding unapproved vendors.
- Timeline: System deployment, data migration, testing, and staff training must be completed within 6 months.
- Data Security: Patient data must be protected through controlled access, secure storage, and authorised user permissions throughout system operation.

13. APPROVALS

Role	Name	Signature	Date
Medical director			
Hospital management			
Operations Director			
Technology Director			
Business Analyst			